Example 2. Solve for the reactions on the beam shown in Fig. 1. Using information on cantilever beams from Table.

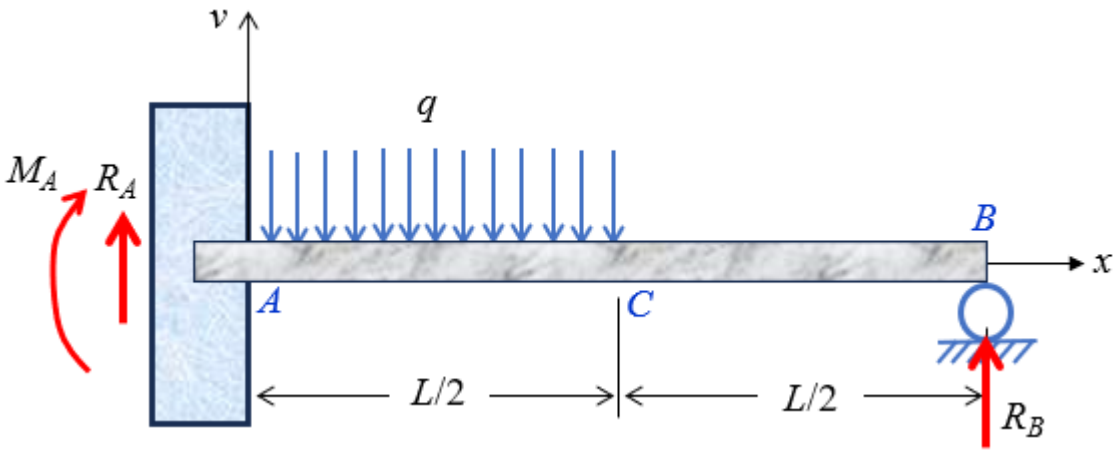


Fig. 1. Original problem

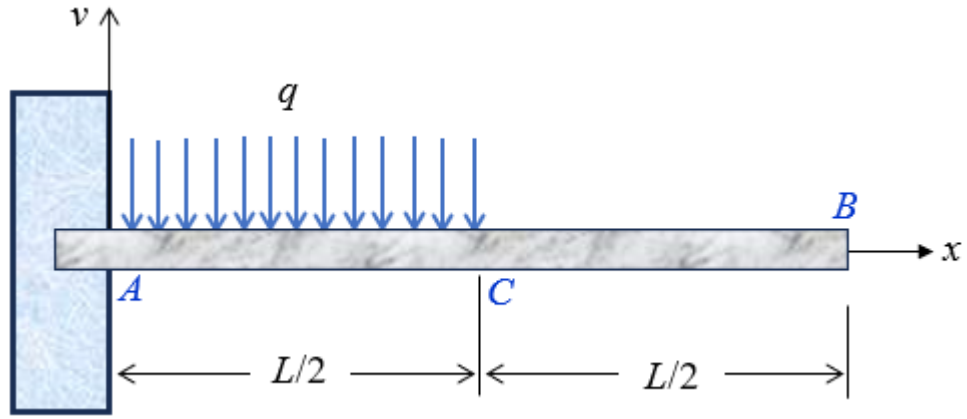


Fig. 2a. Applied load on released structure

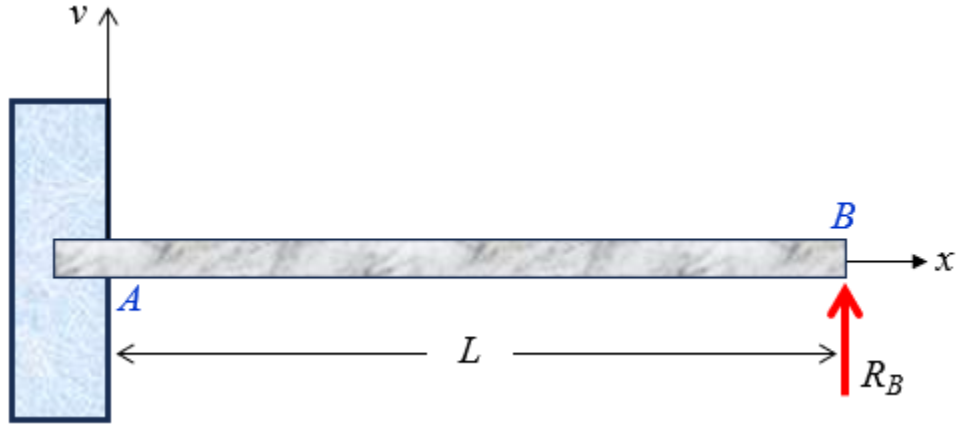


Fig. 2b. Redundant reaction on released structure

Solution

Select the subproblems: We are to select subproblems from the cantilever beam table. This means that R_B is to be considered the redundant reaction. The constituent subproblems are shown in Fig. 2.

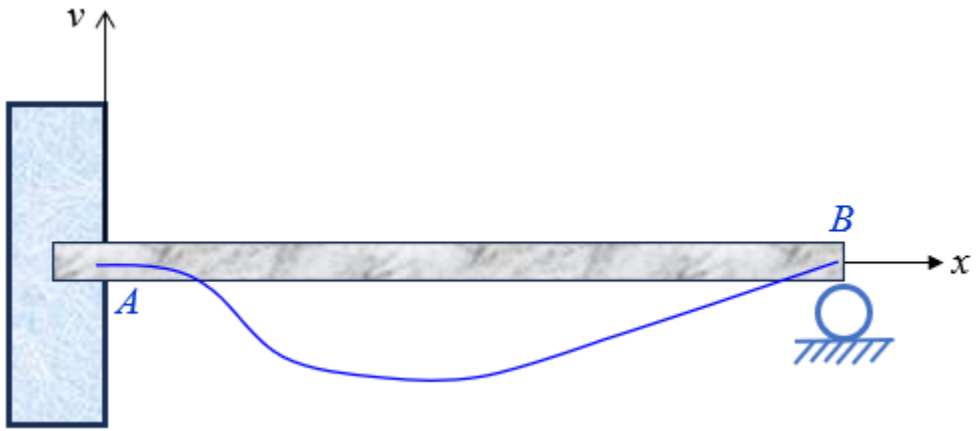
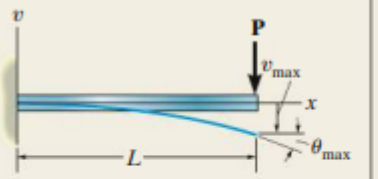
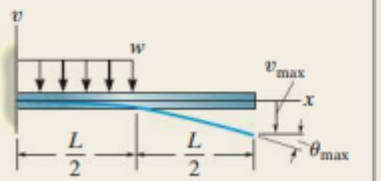


Fig. 3. The expected deflection curve

Table of deflections

Beam	Slope	Deflection	Elastic Curve
	$\theta_{max} = \frac{-PL^2}{2EI}$	$v_{max} = \frac{-PL^3}{3EI}$	$v = \frac{-Px^2}{6EI}(3L - x)$
	$\theta_{max} = \frac{-wL^3}{48EI}$	$v_{max} = \frac{-7wL^4}{384EI}$	$v = \frac{-wx^2}{24EI}\left(x^2 - 2Lx + \frac{3}{2}L^2\right)$ $0 \le x \le L/2$ $v = \frac{-wL^3}{384EI}(8x - L)$ $L/2 \le x \le L$

Sketch the deflection curve: The expected deflection curve is shown in Fig. 3.

Write the superposition equation: The reaction R_B is obtained by writing a constraint equation for the support B. The deflection at point B is zero.

$$v(L) = \delta_a + \delta_b = 0 \qquad (a)$$

Where is δ_a the free-end deflection of beam in Fig. 2a, and δ_b is the free-end deflection of beam in Fig. 2b.

From the table (left), we get

$$\delta_a = -\frac{7qL^4}{384EI}, \quad \delta_b = \frac{R_B L^3}{3EI},$$

$$\text{So,} \quad -\frac{7qL^4}{384EI} + \frac{R_B L^3}{3EI} = 0$$

$$\text{So,} \quad R_B = \frac{7qL}{128}$$

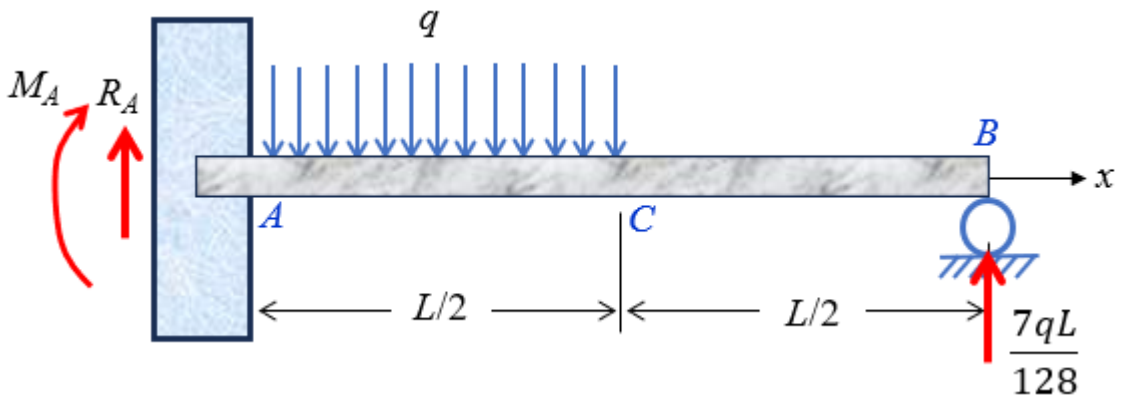


Fig. 1. Original problem

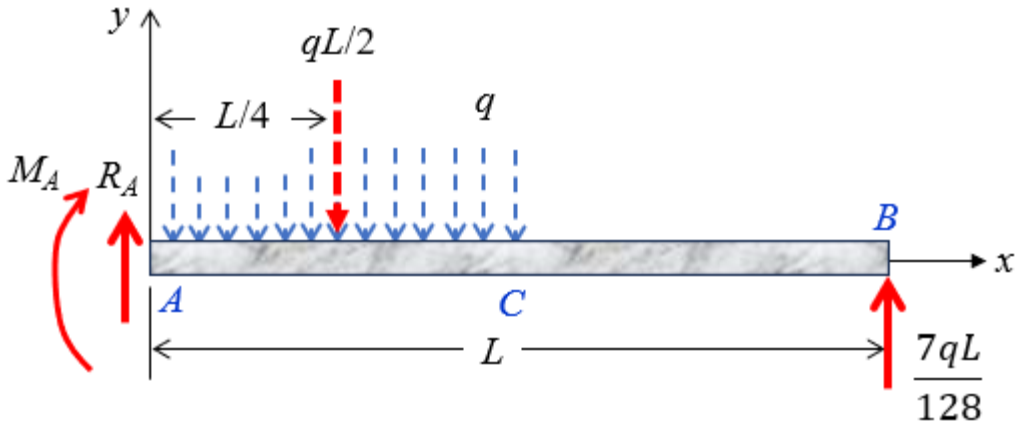


Fig. 4. Free-body diagram

Complete the problem: To solve for the reactions at A we need to study the free-body diagram (Fig. 4) and equilibrium equations.

$$\sum M_{\text{about point } A} = 0, \qquad \left(\frac{7qL}{128}\right)L - \frac{qL}{2}\frac{L}{4} - M_A = 0$$

$$\uparrow \sum F = 0, \qquad \frac{7qL}{128} + R_A - \frac{qL}{2} = 0$$

Therefore, the shear force and bending moment reactions at point A are

$$R_A = \frac{57}{128}qL, \qquad M_A = -\frac{9}{128}qL^2$$

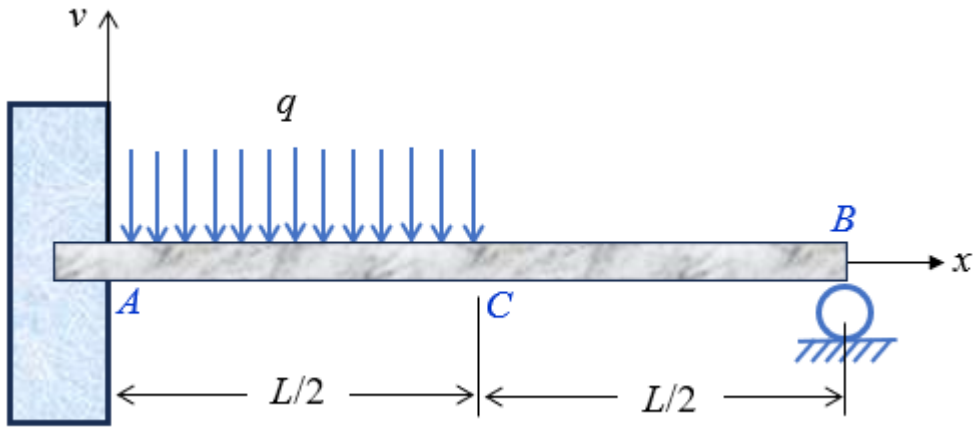


Fig. 1. Original problem

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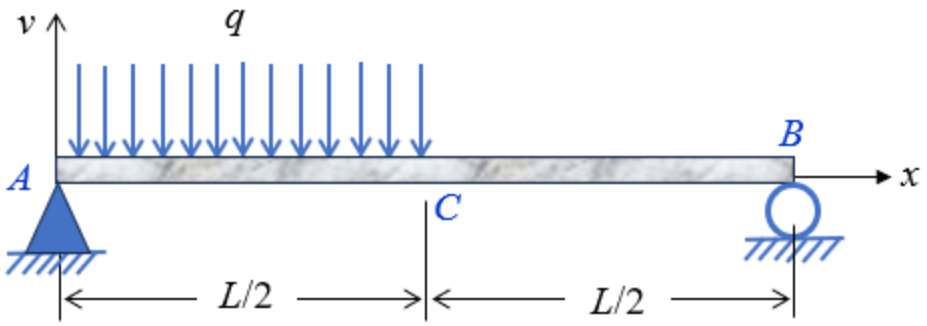


Fig. 2a. Applied load on released structure

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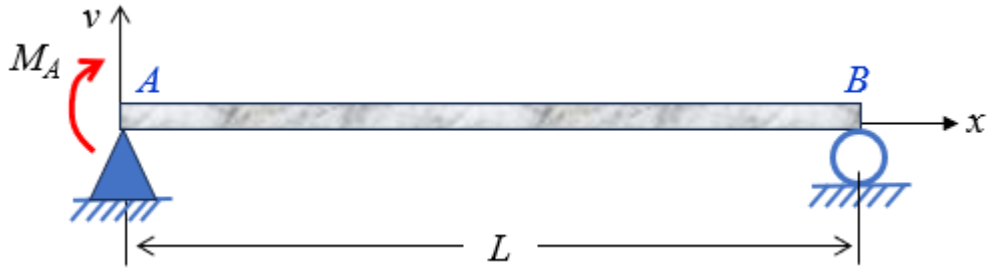


Fig. 2b. Redundant reaction on released structure

Note:

In many cases, there are alternative ways to construct a superposition solution for a statically indeterminate problem. That is, there may be alternative ways to select the redundant(s) to be used, so long as the required solutions are available.

For example, in the problem of Example 1 (Fig. 1), the bending moment reaction M_A at point A can be considered as a redundant, and information for simply supported beams (Fig. 2) from Table can be used.

In planning a superposition solution, you should mentally explore alternatives and try to solve the problem in the most efficient manner.

10.3. Obtaining deflections and reactions by solving moment-curvature equation

In Chapter 8 the second-order ordinary differential equation (moment-curvature equation) governing the deflection of elastic beams was derived. It was used to solve for the deflection and slope of statically determinate beams.

$$\frac{d^2v}{dx^2} = \frac{M}{EI} \qquad (8.1)$$

In this section, we will extend the solution procedure to treat statically indeterminate beam deflection problems.

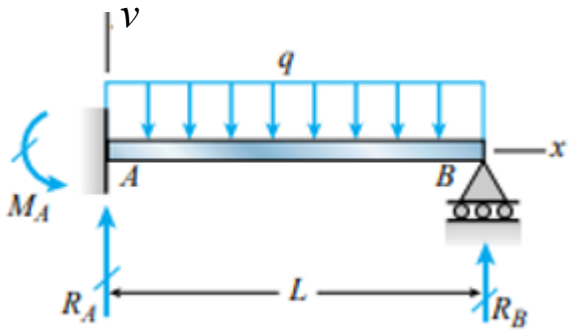
$$\frac{dv}{dx} = \frac{1}{EI}\int M dx + C1 \qquad (8.2)$$

Since the beam is statically indeterminate, the bending moment can not be obtained directly based on the equilibrium equations.

However, a statically indeterminate beam, has more boundary conditions. This additional information will allow us to obtain the reactions and determine the deflection curve.

$$v = \frac{1}{EI}\int\int M dx dx + C_1 x + C_2 \qquad (8.3)$$

Example 3. A propped cantilever beam AB of length L supports a uniform load of intensity q (see below figure). Analyze this beam by solving the second-order differential equation of the deflection curve (the moment-curvature equation). Determine the reactions, shear forces, bending moments, slopes, and deflections of the beam.



Solution

Identify whether the beam is statically determinate or indeterminate: The beam has three unknown reactions (M_A, R_A, and R_B). Only two equilibrium equations are available for determining these reactions, and therefore the beam is statically indeterminate, with one redundant force.

Choose the redundant reaction(s): Let us choose the reaction R_B at the simple support as the redundant.

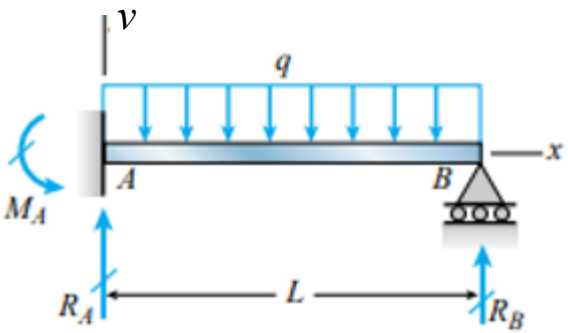
Bending moment in the beam: Use a free-body diagram, the bending moment M at distance x from the fixed support can be expressed in terms of the external distribution q and the redundant force R_B as follows:

$$M = R_B(L - x) - \frac{q(L - x)^2}{2} \qquad \text{(a)}$$

or

$$M = -\frac{qx^2}{2} + (qL - R_B)x + R_B L - \frac{qL^2}{2} \qquad \text{(b)}$$

The above equations (a) and (b) can be obtained by the customary technique of constructing a free-body diagram of part of the beam and solving an equation of equilibrium (Chapter 8).



Differential equation and its solution. The second-order differential equation (Eq 8.1) now becomes

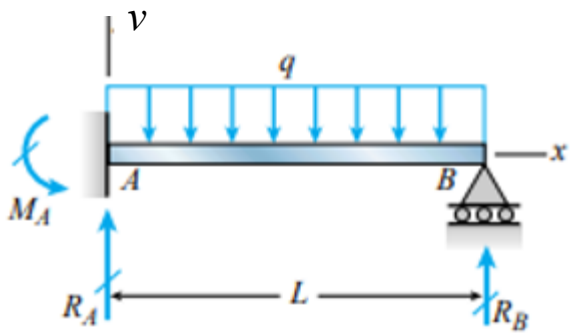
$$EI\frac{\mathrm{d}^2v}{\mathrm{d}x^2} = -\frac{qx^2}{2} + (qL - R_B)x + R_BL - \frac{qL^2}{2} \qquad \text{(c)}$$

After two successive integrations, we obtain the following equations for the slopes and deflections of the beam:

$$EI\frac{\mathrm{d}v}{\mathrm{d}x} = -\frac{q}{6}x^3 + \frac{qL - R_B}{2}x^2 + \left(R_BL - \frac{qL^2}{2}\right)x + C_1 \qquad \text{(d)}$$

$$EIv = -\frac{q}{24}x^4 + \frac{qL - R_B}{6}x^3 + \frac{R_BL - \frac{qL^2}{2}}{2}x^2 + C_1x + C_2 \qquad \text{(e)}$$

These equations contain three unknown quantities (two integration constants C_1, C_2, and the redundant force R_B).



Boundary conditions. Three boundary conditions pertaining to the deflections and slopes of the beam are apparent from an inspection of the figure. These conditions are as follows:
(1) the deflection at the fixed support is zero,
(2) the slope at the fixed support is zero, and
(3) the deflection at the simple support (the roller) is zero. Thus

$$v\Big|_{x=0} = 0, \qquad \frac{dv}{dx}\Big|_{x=0} = 0, \qquad v\Big|_{x=L} = 0 \qquad \text{(f)}$$

Applying these conditions to the equations for slopes and deflections (Eqs. d and e), we find $C_1 = 0$, $C_2 = 0$ and

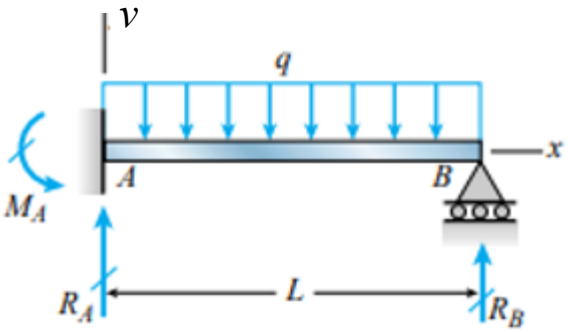
$$R_B = \frac{3qL}{8} \qquad \text{(g)}$$

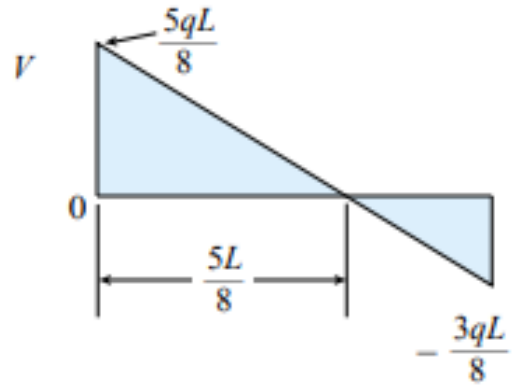
Reactions at the fixed end. By considering the overall equilibrium of the entire beam, we can express the other two reactions:

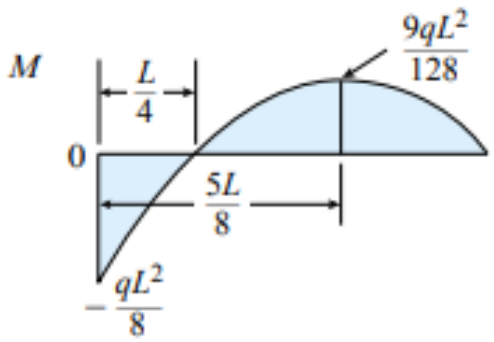
$$R_A = qL - R_B\,, \quad M_A = \frac{qL^2}{2} - R_B L \qquad \text{(h)}$$

So,

$$R_A = \frac{5qL}{8}\,, \quad M_A = \frac{qL^2}{8} \qquad \text{(i)}$$







Shear forces and bending moments. Having obtained the reaction R_B, the bending moment in the beam can be obtained from Eq. (b) as

$$M = \frac{q}{8}(5Lx - L^2 - 4x^2) \qquad \text{(j)}$$

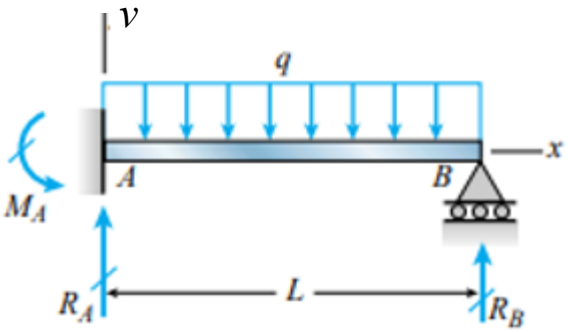
The shear force in the beam can be derived from a free-body diagram, or from the differential equation, Eq. (8.2)

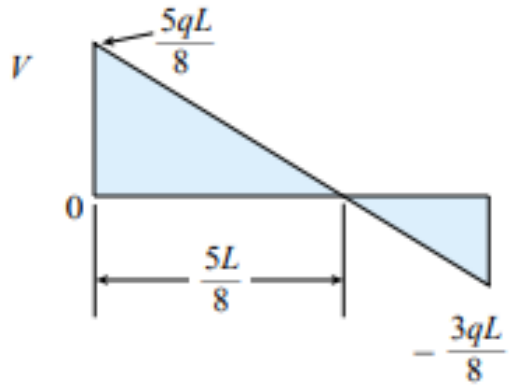
$$V = \frac{dM}{dx} = \frac{q}{8}(5L - 8x) \qquad \text{(k)}$$

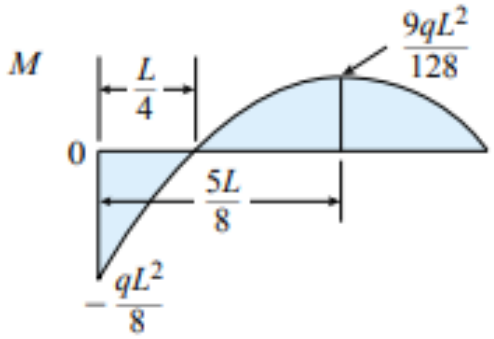
Shear-force and bending-moment diagrams for the beam can be drawn with the aid of these equations (see left figures).

From the diagrams or Eq. (k), we see that the maximum shear force occurs at $x = 0$ and is

(l)







To determine the maximum bending moment, we use $\frac{dM}{dx} = 0$ to find the location of it. From Eq. (j) we get

$$\frac{dM}{dx} = \frac{q}{8}(5L - 8x) \qquad \text{(m)}$$

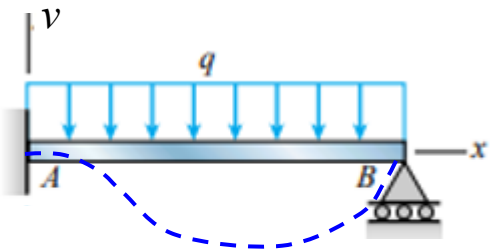
Thus, $\frac{dM}{dx} = 0$ yields $x = \frac{5L}{8}$. Substitute this into Eq. (j) we get

$$M_{\max} = \frac{9qL^2}{128}$$

Slopes and deflections of the beam. Returning to Eqs. (d) and (e) for the slope and deflection, we now substitute the integration constants ($C_1 = 0$, $C_2 = 0$) as well as the redundant R_B (Eq. (g)) and obtain.

$$\frac{dv}{dx} = \frac{q}{48EI}(-8x^3 + 15Lx^2 - 6L^2x) \qquad \text{(n)}$$

$$v = \frac{q}{48EI}(-2x^4 + 5Lx^3 - 3L^2x^2) \qquad \text{(o)}$$



Deflection curve

The deflection curve is shown in the left figure.

To determine the maximum deflection of the beam, we set the slope (Eq. (n)) equal to zero and solve for the distance x to the point where this deflection occurs:

$$\frac{dv}{dx} = 0 \quad \text{or} \quad -8x^3 + 15Lx^2 - 6L^2x = 0 \qquad \text{(r)}$$

From which, we get $x = 0.5785L$.

Substituting this x value into the equation for the deflection, Eq. (o), we get the maximum deflection:

$$v_{\max} = v|_{x=0.5785L} = \frac{qL^4}{184.6EI}$$

Note: In this example, we analyzed the beam by taking the reaction R_B as the redundant reaction. An alternative approach is to take the reactive moment M_A as the redundant. Then we can express the bending moment M in terms of M_A, substitute the resulting expression into the second-order differential equation, and solve as before.